

COLLIE, Australia

WMO: 956210

Lat: 33.3608S

Lon: 116.1717E

Elev: 200

StdP: 98.95

Time Zone: 8.00 (E08)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.3	0.8	-1.5	3.4	1.3	-0.4	3.7	2.7	7.8	16.1	7.0	14.9	0.0	90	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.8	35.1	19.4	33.4	19.1	31.6	18.5	21.5	30.4	20.4	29.5	19.5	28.7	3.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.9	24.7	17.5	12.8	23.1	16.4	11.9	22.4	63.2	30.7	59.3	29.6	56.5	28.7	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.6	5.0	-2.4	38.7	1.1	1.2	-3.2	39.6	-3.8	40.3	-4.4	41.0	-5.3	41.9		
(5)			-2.9	23.3	1.1	1.6	-3.7	24.4	-4.4	25.4	-5.0	26.3	-5.8	27.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	21.9	21.9	19.6	15.9	12.6	10.6	9.8	10.4	11.4	14.1	17.4	19.8	(6)	(7)
	DBStd	5.24	3.12	3.31	3.42	2.91	2.35	2.24	2.22	2.21	2.48	2.91	3.35	3.54	(7)	(8)
	HDD10.0	91	0	0	0	0	4	19	32	21	13	2	0	0	(8)	(9)
	HDD18.3	1472	6	8	27	84	178	233	266	244	209	134	57	26	(9)	(10)
	CDD10.0	2066	368	333	299	177	86	36	24	34	54	129	222	303	(10)	(11)
	CDD18.3	407	116	109	67	11	1	0	0	0	0	4	29	70	(11)	(12)
	CDH23.3	5209	1467	1289	789	156	10	0	0	1	8	90	445	954	(12)	(13)
	CDH26.7	2241	701	590	311	34	1	0	0	0	1	18	161	424	(13)	(14)
(14) Wind	WSAvg	1.9	2.6	2.5	2.2	1.4	1.2	1.4	1.5	1.7	2.1	2.2	2.4	2.3	(14)	(15)
(15) Precipitation	PrecAvg	880	13	14	24	47	119	173	165	128	88	56	35	14	(15)	(16)
	PrecMax	1334	85	80	142	125	270	414	388	257	210	129	106	56	(16)	(17)
	PrecMin	598	0	0	0	0	33	56	64	32	18	2	1	0	(17)	(18)
	PrecStd	177	19	19	30	34	56	76	68	50	44	29	25	13	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.8	37.0	35.8	30.3	24.8	20.4	19.0	20.9	24.5	29.0	33.8	36.6	(19)	(20)
		MCWB	19.4	21.0	18.6	17.6	16.0	13.4	13.1	13.3	14.7	16.7	17.8	19.4	(20)	(21)
	2%	DB	34.9	34.4	32.4	27.2	22.0	18.6	17.3	18.8	21.5	25.7	31.1	34.0	(21)	(22)
		MCWB	19.3	20.1	18.8	16.7	14.5	13.2	12.8	13.3	13.8	16.0	17.4	19.0	(22)	(23)
	5%	DB	33.1	32.5	30.1	24.9	20.0	17.3	16.2	17.2	19.1	23.4	28.3	31.5	(23)	(24)
		MCWB	19.3	19.2	18.4	16.4	14.2	12.9	12.5	12.7	13.3	15.4	16.6	17.9	(24)	(25)
	10%	DB	30.9	30.6	27.8	22.7	18.6	16.0	15.1	15.8	17.3	20.9	25.8	28.7	(25)	(26)
		MCWB	18.6	18.6	17.7	15.8	14.0	12.3	11.9	12.1	12.6	14.2	15.9	17.0	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.6	23.0	21.4	19.7	17.6	15.8	15.2	15.7	16.3	18.4	19.9	21.4	(27)	(28)
		MCDB	30.9	32.1	29.8	25.1	20.5	18.1	17.1	18.3	21.6	25.7	29.0	31.7	(28)	(29)
	2%	WB	21.4	21.6	20.1	18.3	16.3	14.7	14.1	14.4	15.0	17.0	18.5	19.9	(29)	(30)
		MCDB	29.8	30.8	28.7	23.7	19.3	17.0	16.1	17.2	19.4	24.1	26.9	30.9	(30)	(31)
	5%	WB	20.2	20.3	19.2	17.4	15.4	13.7	13.1	13.4	13.9	15.8	17.5	18.8	(31)	(32)
		MCDB	29.4	29.6	27.4	22.4	18.5	16.0	15.4	16.1	17.9	21.9	25.6	28.6	(32)	(33)
	10%	WB	19.1	19.3	18.3	16.5	14.5	12.8	12.2	12.6	13.0	14.8	16.6	17.7	(33)	(34)
		MCDB	28.5	28.6	25.8	21.5	17.7	15.3	14.5	15.3	16.5	20.1	24.4	27.1	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	15.5	15.0	13.9	13.0	11.9	11.2	11.4	11.9	12.9	14.0	15.1	(35)	(36)
		MCDBR	19.7	18.7	18.5	17.6	15.8	13.8	12.7	14.0	15.7	17.0	18.8	19.3	(36)	(37)
	5% WB	MCWBR	6.6	6.5	6.9	8.3	9.4	9.0	8.6	8.8	8.9	8.3	7.1	6.7	(37)	(38)
		MCWBR	15.8	15.7	14.8	13.4	11.6	10.9	9.9	11.5	12.4	14.5	15.5	17.5	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub	taub	0.328	0.327	0.313	0.307	0.302	0.298	0.294	0.305	0.325	0.322	0.322	0.321	(39)	(40)
		taud	2.582	2.593	2.636	2.639	2.620	2.611	2.621	2.574	2.491	2.526	2.555	2.578	(40)	(41)
	Ebn at Noon	Ebn at Noon	1010	991	971	924	876	852	876	911	941	985	1009	1020	(41)	(42)
		Edh at Noon	105	100	90	80	72	68	71	83	102	106	107	106	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg	RadAvg	8.00	7.12	5.73	4.07	2.97	2.42	2.59	3.37	4.53	5.93	7.22	8.03	(43)	(44)
	RadStd	RadStd	0.45	0.28	0.31	0.27	0.22	0.15	0.14	0.19	0.25	0.31	0.45	0.44	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (24 neighbors)	N/A	N/A	N/A	+0.43	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page